```
(medai) D:\Research Paper Work\Multi Organ Cancer Detector\src>python preextract.py

======================================================================
  PATCH PRE-EXTRACTION (for 3-fold cross-validation)
  Run this ONCE. Then run main.py for training.
======================================================================

  Found 888 CT scans in D:\Research Paper Work\Multi Organ Cancer Detector\data\LUNA16
  Resampling to:   1x1x1 mm isotropic spacing
  Candidates with matching scans: 551065 (1351 nodule, 549714 non-nodule)

  After balancing (neg_ratio=5):
    Positives (nodule):     1351
    Negatives (non-nodule): 6755
    Ratio:                  1:5
    Unique patients:        887

  Saving patches to: D:\Research Paper Work\Multi Organ Cancer Detector\data\patches

  1/887 scans | 9 patches | 1s elapsed | ~1285s remaining
  50/887 scans | 423 patches | 126s elapsed | ~2117s remaining
  100/887 scans | 901 patches | 264s elapsed | ~2082s remaining
  150/887 scans | 1371 patches | 392s elapsed | ~1925s remaining
  200/887 scans | 1831 patches | 524s elapsed | ~1800s remaining
  250/887 scans | 2345 patches | 611s elapsed | ~1558s remaining
  300/887 scans | 2784 patches | 692s elapsed | ~1353s remaining
  350/887 scans | 3227 patches | 769s elapsed | ~1180s remaining
  400/887 scans | 3656 patches | 853s elapsed | ~1039s remaining
  450/887 scans | 4067 patches | 934s elapsed | ~907s remaining
  500/887 scans | 4523 patches | 1014s elapsed | ~785s remaining
  550/887 scans | 5001 patches | 1100s elapsed | ~674s remaining
  600/887 scans | 5441 patches | 1179s elapsed | ~564s remaining
  650/887 scans | 5899 patches | 1261s elapsed | ~460s remaining
  700/887 scans | 6344 patches | 1342s elapsed | ~358s remaining
  750/887 scans | 6847 patches | 1426s elapsed | ~260s remaining
  800/887 scans | 7298 patches | 1506s elapsed | ~164s remaining
  850/887 scans | 7771 patches | 1584s elapsed | ~69s remaining
  887/887 scans | 8106 patches | 1642s elapsed | ~0s remaining
```

```
  EXTRACTION COMPLETE
  Total patches:  8106 (1351 nodule, 6755 non-nodule)
  Manifest:       D:\Research Paper Work\Multi Organ Cancer Detector\data\patches\manifest.csv
  Disk usage:     ~8.5 GB
  Total time:     27.4 minutes

  Next step:  python main.py
======================================================================


(medai) D:\Research Paper Work\Multi Organ Cancer Detector\src>python main.py

================================================================================
  LUNA16 NODULE CLASSIFICATION
  MULTI-MODEL 3-FOLD CROSS-VALIDATION PIPELINE
================================================================================

STEP 1: READING MANIFEST
--------------------------------------------------------------------------------
  Total patches:   8106 (1351 nodule, 6755 non-nodule)
  Unique patients: 887

STEP 2: SETTING UP 3-FOLD CROSS-VALIDATION
--------------------------------------------------------------------------------
  Patient-level splitting (prevents data leakage):

    Fold 1: 295 patients, 2711 patches (430 nodule, 2281 non-nodule)
    Fold 2: 295 patients, 2720 patches (471 nodule, 2249 non-nodule)
    Fold 3: 297 patients, 2675 patches (450 nodule, 2225 non-nodule)

  Device: cuda
  GPU:    NVIDIA GeForce RTX 4050 Laptop GPU
  VRAM:   6.4 GB

  Models to train: ResNet3D, VGG3D


********************************************************************************
  TRAINING MODEL 1/2: ResNet3D
  Pipeline elapsed: 0 min
```

```
################################################################################
#  MODEL: ResNet3D (resnet3d)
################################################################################

================================================================================
  ResNet3D - FOLD 1 / 3
================================================================================

  Split: 504 train patients, 88 val patients, 295 test patients
  Datasets:
    train:  4604 patches (794 nodule, 3810 non-nodule, ratio 1:4)
    val  :   791 patches (127 nodule, 664 non-nodule, ratio 1:5)
    test :  2711 patches (430 nodule, 2281 non-nodule, ratio 1:5)
  Parameters: 3.62M (3,615,970 total)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1788 TrA:0.514 | VaL:0.1708 VaA:0.592 | LR:6.2e-05 | 117s/ep | fold:2m total:2m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1634 TrA:0.621 | VaL:0.1612 VaA:0.635 | LR:3.0e-04 | 103s/ep | fold:9m total:9m
    Ep  10/100 | TrL:0.1562 TrA:0.667 | VaL:0.1482 VaA:0.671 | LR:3.0e-04 | 103s/ep | fold:17m total:17m
    Ep  15/100 | TrL:0.1528 TrA:0.683 | VaL:0.1955 VaA:0.469 | LR:2.9e-04 | 140s/ep | fold:28m total:28m
    Ep  20/100 | TrL:0.1243 TrA:0.767 | VaL:0.1138 VaA:0.777 | LR:2.8e-04 | 103s/ep | fold:37m total:37m
    Ep  25/100 | TrL:0.0416 TrA:0.944 | VaL:0.0306 VaA:0.956 | LR:2.7e-04 | 103s/ep | fold:45m total:45m
    Ep  30/100 | TrL:0.0310 TrA:0.963 | VaL:0.0353 VaA:0.961 | LR:2.5e-04 | 103s/ep | fold:54m total:54m
    Ep  35/100 | TrL:0.0256 TrA:0.965 | VaL:0.0265 VaA:0.957 | LR:2.3e-04 | 103s/ep | fold:62m total:62m
    Ep  40/100 | TrL:0.0201 TrA:0.977 | VaL:0.0298 VaA:0.954 | LR:2.1e-04 | 102s/ep | fold:72m total:72m
    Ep  45/100 | TrL:0.0162 TrA:0.982 | VaL:0.0322 VaA:0.956 | LR:1.9e-04 | 102s/ep | fold:80m total:80m
    Ep  50/100 | TrL:0.0133 TrA:0.981 | VaL:0.0375 VaA:0.962 | LR:1.6e-04 | 102s/ep | fold:89m total:89m
    Ep  55/100 | TrL:0.0138 TrA:0.982 | VaL:0.0322 VaA:0.960 | LR:1.4e-04 | 102s/ep | fold:97m total:97m
    Ep  60/100 | TrL:0.0115 TrA:0.987 | VaL:0.0360 VaA:0.957 | LR:1.1e-04 | 102s/ep | fold:106m total:106m
    Ep  65/100 | TrL:0.0093 TrA:0.987 | VaL:0.0359 VaA:0.954 | LR:9.0e-05 | 102s/ep | fold:114m total:114m
    Ep  70/100 | TrL:0.0089 TrA:0.988 | VaL:0.0430 VaA:0.966 | LR:6.8e-05 | 102s/ep | fold:123m total:123m
    Ep  75/100 | TrL:0.0073 TrA:0.990 | VaL:0.0340 VaA:0.967 | LR:4.8e-05 | 102s/ep | fold:131m total:131m
    Ep  80/100 | TrL:0.0057 TrA:0.992 | VaL:0.0364 VaA:0.971 | LR:3.2e-05 | 102s/ep | fold:140m total:140m

    Early stopping at epoch 81 (no improvement for 25 epochs)
```

```
  ResNet3D Fold 1 training complete: 81 epochs in 141.6 min (best val loss: 0.0237)
  Estimated time remaining for ResNet3D: ~283 min

  Evaluating ResNet3D fold 1 on test set (2711 samples) with TTA...

============================================================
  EVALUATION METRICS - ResNet3D FOLD 1 TEST
============================================================
  Sensitivity (Recall):  0.9326
  Specificity:           0.9847
  Precision:             0.9197
  Accuracy:              0.9764
  F1 Score:              0.9261
  AUC-ROC:               0.9965

  Confusion Matrix:
    True Positives:   401  (Caught nodules)
    True Negatives:  2246  (Correct normals)
    False Positives:   35  (False alarms)
    False Negatives:   29  (Missed nodules)
============================================================


  Generating Grad-CAM visualizations for ResNet3D...
    TP: 3 figure(s) saved
    TN: 3 figure(s) saved
    FP: 3 figure(s) saved
    FN: 3 figure(s) saved
  Grad-CAM figures saved to: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\gradcam\resnet3d
```

```
==============================================================================
  ResNet3D - FOLD 2 / 3
==============================================================================

  Split: 504 train patients, 88 val patients, 295 test patients
  Datasets:
    train:  4644 patches (746 nodule, 3898 non-nodule, ratio 1:5)
    val  :   742 patches (134 nodule, 608 non-nodule, ratio 1:4)
    test :  2720 patches (471 nodule, 2249 non-nodule, ratio 1:4)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1797 TrA:0.510 | VaL:0.1695 VaA:0.598 | LR:6.2e-05 | 116s/ep | fold:2m total:146m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1643 TrA:0.618 | VaL:0.1259 VaA:0.779 | LR:3.0e-04 | 104s/ep | fold:9m total:153m
    Ep  10/100 | TrL:0.1580 TrA:0.648 | VaL:0.1549 VaA:0.648 | LR:3.0e-04 | 103s/ep | fold:17m total:161m
    Ep  15/100 | TrL:0.1439 TrA:0.711 | VaL:0.1719 VaA:0.563 | LR:2.9e-04 | 103s/ep | fold:26m total:170m
    Ep  20/100 | TrL:0.0520 TrA:0.927 | VaL:0.0422 VaA:0.942 | LR:2.8e-04 | 103s/ep | fold:35m total:179m
    Ep  25/100 | TrL:0.0307 TrA:0.963 | VaL:0.0319 VaA:0.966 | LR:2.7e-04 | 103s/ep | fold:43m total:187m
    Ep  30/100 | TrL:0.0284 TrA:0.961 | VaL:0.0480 VaA:0.945 | LR:2.5e-04 | 103s/ep | fold:52m total:196m
    Ep  35/100 | TrL:0.0194 TrA:0.976 | VaL:0.0458 VaA:0.956 | LR:2.3e-04 | 103s/ep | fold:60m total:204m
    Ep  40/100 | TrL:0.0189 TrA:0.977 | VaL:0.0321 VaA:0.970 | LR:2.1e-04 | 103s/ep | fold:69m total:213m
    Ep  45/100 | TrL:0.0165 TrA:0.982 | VaL:0.0330 VaA:0.974 | LR:1.9e-04 | 103s/ep | fold:78m total:222m
    Ep  50/100 | TrL:0.0141 TrA:0.982 | VaL:0.0395 VaA:0.974 | LR:1.6e-04 | 103s/ep | fold:86m total:230m
    Ep  55/100 | TrL:0.0107 TrA:0.986 | VaL:0.0279 VaA:0.972 | LR:1.4e-04 | 103s/ep | fold:95m total:239m
    Ep  60/100 | TrL:0.0124 TrA:0.986 | VaL:0.0303 VaA:0.977 | LR:1.1e-04 | 103s/ep | fold:103m total:247m
    Ep  65/100 | TrL:0.0098 TrA:0.988 | VaL:0.0321 VaA:0.977 | LR:9.0e-05 | 103s/ep | fold:112m total:256m
    Ep  70/100 | TrL:0.0059 TrA:0.991 | VaL:0.0557 VaA:0.973 | LR:6.8e-05 | 103s/ep | fold:120m total:264m
    Ep  75/100 | TrL:0.0054 TrA:0.993 | VaL:0.0390 VaA:0.982 | LR:4.8e-05 | 103s/ep | fold:129m total:273m

    Early stopping at epoch 76 (no improvement for 25 epochs)

  ResNet3D Fold 2 training complete: 76 epochs in 130.6 min (best val loss: 0.0171)
  Estimated time remaining for ResNet3D: ~136 min
```

```
  Evaluating ResNet3D fold 2 on test set (2720 samples) with TTA...

============================================================
  EVALUATION METRICS - ResNet3D FOLD 2 TEST
============================================================
  Sensitivity (Recall):  0.9342
  Specificity:           0.9778
  Precision:             0.8980
  Accuracy:              0.9702
  F1 Score:              0.9157
  AUC-ROC:               0.9955

  Confusion Matrix:
    True Positives:   440  (Caught nodules)
    True Negatives:  2199  (Correct normals)
    False Positives:   50  (False alarms)
    False Negatives:   31  (Missed nodules)
============================================================
```

```
================================================================================
  ResNet3D - FOLD 3 / 3
================================================================================

  Split: 502 train patients, 88 val patients, 297 test patients
  Datasets:
    train:  4597 patches (768 nodule, 3829 non-nodule, ratio 1:4)
    val  :   834 patches (133 nodule, 701 non-nodule, ratio 1:5)
    test :  2675 patches (450 nodule, 2225 non-nodule, ratio 1:4)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1780 TrA:0.505 | VaL:0.1770 VaA:0.361 | LR:6.2e-05 | 117s/ep | fold:2m total:279m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1614 TrA:0.641 | VaL:0.1488 VaA:0.737 | LR:3.0e-04 | 103s/ep | fold:9m total:286m
    Ep  10/100 | TrL:0.1573 TrA:0.651 | VaL:0.1551 VaA:0.698 | LR:3.0e-04 | 103s/ep | fold:17m total:294m
    Ep  15/100 | TrL:0.1493 TrA:0.682 | VaL:0.1903 VaA:0.489 | LR:2.9e-04 | 103s/ep | fold:26m total:303m
    Ep  20/100 | TrL:0.1381 TrA:0.724 | VaL:0.1420 VaA:0.716 | LR:2.8e-04 | 103s/ep | fold:35m total:312m
    Ep  25/100 | TrL:0.1312 TrA:0.747 | VaL:0.1327 VaA:0.725 | LR:2.7e-04 | 103s/ep | fold:43m total:320m
    Ep  30/100 | TrL:0.0877 TrA:0.863 | VaL:0.0898 VaA:0.865 | LR:2.5e-04 | 103s/ep | fold:52m total:329m
    Ep  35/100 | TrL:0.0363 TrA:0.952 | VaL:0.0269 VaA:0.964 | LR:2.3e-04 | 103s/ep | fold:60m total:337m
    Ep  40/100 | TrL:0.0268 TrA:0.963 | VaL:0.0494 VaA:0.944 | LR:2.1e-04 | 103s/ep | fold:69m total:346m
    Ep  45/100 | TrL:0.0196 TrA:0.977 | VaL:0.0221 VaA:0.975 | LR:1.9e-04 | 103s/ep | fold:78m total:354m
    Ep  50/100 | TrL:0.0163 TrA:0.981 | VaL:0.0297 VaA:0.969 | LR:1.6e-04 | 103s/ep | fold:86m total:363m
    Ep  55/100 | TrL:0.0157 TrA:0.982 | VaL:0.0274 VaA:0.964 | LR:1.4e-04 | 102s/ep | fold:95m total:372m
    Ep  60/100 | TrL:0.0114 TrA:0.986 | VaL:0.0221 VaA:0.977 | LR:1.1e-04 | 102s/ep | fold:103m total:380m
    Ep  65/100 | TrL:0.0068 TrA:0.990 | VaL:0.0201 VaA:0.981 | LR:9.0e-05 | 103s/ep | fold:112m total:389m

    Early stopping at epoch 69 (no improvement for 25 epochs)

  ResNet3D Fold 3 training complete: 69 epochs in 118.6 min (best val loss: 0.0162)

  Evaluating ResNet3D fold 3 on test set (2675 samples) with TTA...
```

```
============================================================
  EVALUATION METRICS - ResNet3D FOLD 3 TEST
============================================================
  Sensitivity (Recall):  0.8556
  Specificity:           0.9825
  Precision:             0.9080
  Accuracy:              0.9611
  F1 Score:              0.8810
  AUC-ROC:               0.9834

  Confusion Matrix:
    True Positives:   385  (Caught nodules)
    True Negatives:  2186  (Correct normals)
    False Positives:   39  (False alarms)
    False Negatives:   65  (Missed nodules)
============================================================


================================================================================
  ResNet3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  -------------------------------------------------------
  sensitivity       0.9326   0.9342   0.8556 0.9074+/-0.0367
  specificity       0.9847   0.9778   0.9825 0.9816+/-0.0029
  precision         0.9197   0.8980   0.9080 0.9086+/-0.0089
  accuracy          0.9764   0.9702   0.9611 0.9692+/-0.0063
  auc_roc           0.9965   0.9955   0.9834 0.9918+/-0.0059
  f1                0.9261   0.9157   0.8810 0.9076+/-0.0193

  Aggregate (all folds combined, 8106 samples):
    sensitivity     0.9075
    specificity     0.9816
    precision       0.9081
    accuracy        0.9693
    auc_roc         0.9925
    f1              0.9078
```

```
  ROC curve saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\resnet3d\roc_curve_kfold.png
  Training curves saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\resnet3d\training_curves_kfold.png

******************************************************************************
  TRAINING MODEL 2/2: VGG3D
  Pipeline elapsed: 398 min
******************************************************************************

##############################################################################
#  MODEL: VGG3D (vgg3d)
##############################################################################

==============================================================================
  VGG3D - FOLD 1 / 3
==============================================================================

  Split: 504 train patients, 88 val patients, 295 test patients
  Datasets:
    train:  4604 patches (794 nodule, 3810 non-nodule, ratio 1:4)
    val  :   791 patches (127 nodule, 664 non-nodule, ratio 1:5)
    test :  2711 patches (430 nodule, 2281 non-nodule, ratio 1:5)
  Parameters: 0.87M (866,338 total)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1866 TrA:0.509 | VaL:0.1762 VaA:0.397 | LR:6.2e-05 | 78s/ep | fold:1m total:399m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1659 TrA:0.606 | VaL:0.1651 VaA:0.623 | LR:3.0e-04 | 64s/ep | fold:6m total:403m
    Ep  10/100 | TrL:0.1499 TrA:0.695 | VaL:0.1450 VaA:0.719 | LR:3.0e-04 | 64s/ep | fold:11m total:409m
    Ep  15/100 | TrL:0.0632 TrA:0.925 | VaL:0.0542 VaA:0.932 | LR:2.9e-04 | 64s/ep | fold:16m total:414m
    Ep  20/100 | TrL:0.0415 TrA:0.944 | VaL:0.0489 VaA:0.933 | LR:2.8e-04 | 64s/ep | fold:22m total:419m
    Ep  25/100 | TrL:0.0294 TrA:0.964 | VaL:0.0422 VaA:0.938 | LR:2.7e-04 | 64s/ep | fold:27m total:425m
    Ep  30/100 | TrL:0.0290 TrA:0.964 | VaL:0.0372 VaA:0.951 | LR:2.5e-04 | 64s/ep | fold:32m total:430m
    Ep  35/100 | TrL:0.0240 TrA:0.967 | VaL:0.0312 VaA:0.948 | LR:2.3e-04 | 64s/ep | fold:38m total:436m
    Ep  40/100 | TrL:0.0221 TrA:0.974 | VaL:0.0268 VaA:0.957 | LR:2.1e-04 | 64s/ep | fold:43m total:441m
    Ep  45/100 | TrL:0.0175 TrA:0.980 | VaL:0.0270 VaA:0.957 | LR:1.9e-04 | 64s/ep | fold:48m total:446m
```

```
    Ep  50/100 | TrL:0.0140 TrA:0.981 | VaL:0.0451 VaA:0.943 | LR:1.6e-04 | 64s/ep | fold:54m total:452m

    Early stopping at epoch 54 (no improvement for 25 epochs)

  VGG3D Fold 1 training complete: 54 epochs in 58.0 min (best val loss: 0.0204)
  Estimated time remaining for VGG3D: ~116 min

  Evaluating VGG3D fold 1 on test set (2711 samples) with TTA...

============================================================
  EVALUATION METRICS - VGG3D FOLD 1 TEST
============================================================
  Sensitivity (Recall):  0.9326
  Specificity:           0.9851
  Precision:             0.9218
  Accuracy:              0.9768
  F1 Score:              0.9272
  AUC-ROC:               0.9949

  Confusion Matrix:
    True Positives:   401  (Caught nodules)
    True Negatives:  2247  (Correct normals)
    False Positives:   34  (False alarms)
    False Negatives:   29  (Missed nodules)
============================================================


  Generating Grad-CAM visualizations for VGG3D...
    TP: 3 figure(s) saved
    TN: 3 figure(s) saved
    FP: 3 figure(s) saved
    FN: 3 figure(s) saved
  Grad-CAM figures saved to: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\gradcam\vgg3d
```

```
================================================================================
  VGG3D - FOLD 2 / 3
================================================================================

  Split: 504 train patients, 88 val patients, 295 test patients
  Datasets:
    train:  4644 patches (746 nodule, 3898 non-nodule, ratio 1:5)
    val  :   742 patches (134 nodule, 608 non-nodule, ratio 1:4)
    test :  2720 patches (471 nodule, 2249 non-nodule, ratio 1:4)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.2084 TrA:0.506 | VaL:0.1753 VaA:0.453 | LR:6.2e-05 | 77s/ep | fold:1m total:459m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1645 TrA:0.614 | VaL:0.1662 VaA:0.594 | LR:3.0e-04 | 64s/ep | fold:6m total:463m
    Ep  10/100 | TrL:0.1405 TrA:0.720 | VaL:0.1252 VaA:0.790 | LR:3.0e-04 | 64s/ep | fold:11m total:468m
    Ep  15/100 | TrL:0.0610 TrA:0.916 | VaL:0.0654 VaA:0.933 | LR:2.9e-04 | 64s/ep | fold:16m total:474m
    Ep  20/100 | TrL:0.0398 TrA:0.947 | VaL:0.0322 VaA:0.964 | LR:2.8e-04 | 64s/ep | fold:22m total:479m
    Ep  25/100 | TrL:0.0333 TrA:0.961 | VaL:0.0264 VaA:0.970 | LR:2.7e-04 | 64s/ep | fold:27m total:484m
    Ep  30/100 | TrL:0.0259 TrA:0.967 | VaL:0.0295 VaA:0.970 | LR:2.5e-04 | 64s/ep | fold:32m total:490m
    Ep  35/100 | TrL:0.0196 TrA:0.977 | VaL:0.0273 VaA:0.969 | LR:2.3e-04 | 64s/ep | fold:38m total:495m
    Ep  40/100 | TrL:0.0187 TrA:0.975 | VaL:0.0452 VaA:0.953 | LR:2.1e-04 | 64s/ep | fold:43m total:501m
    Ep  45/100 | TrL:0.0164 TrA:0.979 | VaL:0.0326 VaA:0.969 | LR:1.9e-04 | 64s/ep | fold:49m total:506m
    Ep  50/100 | TrL:0.0156 TrA:0.980 | VaL:0.0273 VaA:0.972 | LR:1.6e-04 | 64s/ep | fold:54m total:511m
    Ep  55/100 | TrL:0.0128 TrA:0.981 | VaL:0.0415 VaA:0.960 | LR:1.4e-04 | 64s/ep | fold:59m total:517m
    Ep  60/100 | TrL:0.0147 TrA:0.983 | VaL:0.0296 VaA:0.977 | LR:1.1e-04 | 64s/ep | fold:65m total:522m
    Ep  65/100 | TrL:0.0116 TrA:0.986 | VaL:0.0330 VaA:0.973 | LR:9.0e-05 | 64s/ep | fold:70m total:527m

    Early stopping at epoch 67 (no improvement for 25 epochs)

  VGG3D Fold 2 training complete: 67 epochs in 72.1 min (best val loss: 0.0230)
  Estimated time remaining for VGG3D: ~65 min

  Evaluating VGG3D fold 2 on test set (2720 samples) with TTA...
```

```
============================================================
  EVALUATION METRICS - VGG3D FOLD 2 TEST
============================================================
  Sensitivity (Recall):  0.9427
  Specificity:           0.9809
  Precision:             0.9117
  Accuracy:              0.9743
  F1 Score:              0.9269
  AUC-ROC:               0.9968

  Confusion Matrix:
    True Positives:   444  (Caught nodules)
    True Negatives:  2206  (Correct normals)
    False Positives:   43  (False alarms)
    False Negatives:   27  (Missed nodules)
============================================================
```

```
================================================================================
  VGG3D - FOLD 3 / 3
================================================================================

  Split: 502 train patients, 88 val patients, 297 test patients
  Datasets:
    train:  4597 patches (768 nodule, 3829 non-nodule, ratio 1:4)
    val  :   834 patches (133 nodule, 701 non-nodule, ratio 1:5)
    test :  2675 patches (450 nodule, 2225 non-nodule, ratio 1:4)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1847 TrA:0.503 | VaL:0.1716 VaA:0.534 | LR:6.2e-05 | 77s/ep | fold:1m total:532m
D:\Softwares\Miniconda\envs\medai\lib\site-packages\torch\optim\lr_scheduler.py:240: UserWarning: The epoch parameter in `scheduler.step()` was not necessary and is being deprecated where p
ossible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the closed form is used instead of the new chainable form, where avail
able. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choose.
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.1627 TrA:0.631 | VaL:0.1536 VaA:0.664 | LR:3.0e-04 | 64s/ep | fold:6m total:536m
    Ep  10/100 | TrL:0.1570 TrA:0.655 | VaL:0.1519 VaA:0.656 | LR:3.0e-04 | 64s/ep | fold:11m total:542m
    Ep  15/100 | TrL:0.1409 TrA:0.721 | VaL:0.1350 VaA:0.737 | LR:2.9e-04 | 64s/ep | fold:16m total:547m
    Ep  20/100 | TrL:0.0611 TrA:0.915 | VaL:0.0816 VaA:0.905 | LR:2.8e-04 | 64s/ep | fold:22m total:553m
    Ep  25/100 | TrL:0.0360 TrA:0.954 | VaL:0.0587 VaA:0.938 | LR:2.7e-04 | 64s/ep | fold:27m total:558m
    Ep  30/100 | TrL:0.0287 TrA:0.963 | VaL:0.0427 VaA:0.953 | LR:2.5e-04 | 64s/ep | fold:32m total:563m
    Ep  35/100 | TrL:0.0284 TrA:0.965 | VaL:0.0301 VaA:0.966 | LR:2.3e-04 | 64s/ep | fold:38m total:569m
    Ep  40/100 | TrL:0.0212 TrA:0.975 | VaL:0.0427 VaA:0.946 | LR:2.1e-04 | 64s/ep | fold:43m total:574m
    Ep  45/100 | TrL:0.0224 TrA:0.975 | VaL:0.0328 VaA:0.957 | LR:1.9e-04 | 64s/ep | fold:48m total:579m
    Ep  50/100 | TrL:0.0153 TrA:0.982 | VaL:0.0227 VaA:0.974 | LR:1.6e-04 | 64s/ep | fold:54m total:585m
    Ep  55/100 | TrL:0.0137 TrA:0.986 | VaL:0.0359 VaA:0.962 | LR:1.4e-04 | 64s/ep | fold:59m total:590m
    Ep  60/100 | TrL:0.0115 TrA:0.988 | VaL:0.0327 VaA:0.966 | LR:1.1e-04 | 64s/ep | fold:64m total:595m
    Ep  65/100 | TrL:0.0121 TrA:0.985 | VaL:0.0370 VaA:0.966 | LR:9.0e-05 | 64s/ep | fold:70m total:601m
    Ep  70/100 | TrL:0.0084 TrA:0.991 | VaL:0.0328 VaA:0.976 | LR:6.8e-05 | 64s/ep | fold:75m total:606m
    Ep  75/100 | TrL:0.0082 TrA:0.991 | VaL:0.0343 VaA:0.974 | LR:4.8e-05 | 64s/ep | fold:80m total:611m
    Ep  80/100 | TrL:0.0083 TrA:0.992 | VaL:0.0343 VaA:0.977 | LR:3.2e-05 | 64s/ep | fold:86m total:617m

    Early stopping at epoch 81 (no improvement for 25 epochs)

  VGG3D Fold 3 training complete: 81 epochs in 86.8 min (best val loss: 0.0222)

  Evaluating VGG3D fold 3 on test set (2675 samples) with TTA...
```

```
============================================================
  EVALUATION METRICS - VGG3D FOLD 3 TEST
============================================================
  Sensitivity (Recall):  0.8978
  Specificity:           0.9834
  Precision:             0.9161
  Accuracy:              0.9690
  F1 Score:              0.9068
  AUC-ROC:               0.9917

  Confusion Matrix:
    True Positives:   404  (Caught nodules)
    True Negatives:  2188  (Correct normals)
    False Positives:   37  (False alarms)
    False Negatives:   46  (Missed nodules)
============================================================
```

```
  VGG3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  -------------------------------------------------------
  sensitivity       0.9326   0.9427   0.8978 0.9243+/-0.0192
  specificity       0.9851   0.9809   0.9834 0.9831+/-0.0017
  precision         0.9218   0.9117   0.9161 0.9165+/-0.0041
  accuracy          0.9768   0.9743   0.9690 0.9733+/-0.0032
  auc_roc           0.9949   0.9968   0.9917 0.9944+/-0.0021
  f1                0.9272   0.9269   0.9068 0.9203+/-0.0095

  Aggregate (all folds combined, 8106 samples):
    sensitivity     0.9245
    specificity     0.9831
    precision       0.9164
    accuracy        0.9734
    auc_roc         0.9936
    f1              0.9204

  ROC curve saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\vgg3d\roc_curve_kfold.png
  Training curves saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\vgg3d\training_curves_kfold.png

================================================================================
  STEP 4: CROSS-MODEL COMPARISON
================================================================================

  Model           Params     sensitivity     specificity     precision       accuracy        auc_roc         f1              Time
  ---------------------------------------------------------------------------------------------------------------------------------
  ResNet3D        3.62M      0.907+/-0.037  0.982+/-0.003  0.909+/-0.009  0.969+/-0.006  0.992+/-0.006  0.908+/-0.019  398m
  VGG3D           0.87M      0.924+/-0.019  0.983+/-0.002  0.917+/-0.004  0.973+/-0.003  0.994+/-0.002  0.920+/-0.010  221m

  Comparison ROC saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\comparison_roc.png
  Results summary saved: D:\Research Paper Work\Multi Organ Cancer Detector\results\plots\results_summary.txt
```

```
METHODOLOGY:
  Validation:      3-fold cross-validation
  Split strategy:  Patient-level (prevents data leakage)
  Total patches:   8106 (1351 nodule, 6755 non-nodule)
  Unique patients: 887
  Class balance:   WeightedRandomSampler (50/50 in each batch)
  Val fraction:    15% of non-test patients
  Loss function:   Focal Loss (gamma=2.0)
  Augmentation:    3D flips (all axes), 90-deg rotation (all planes), Gaussian noise, intensity shift/scale
  Test-time aug:   7 geometric transforms averaged

MODEL: ResNet3D
  Architecture:    ResNet3D
  Parameters:      3.62M (3,615,970)
  Input shape:     (1, 64, 64, 64) at 1mm isotropic
  Optimizer:       AdamW (lr=0.0003, wd=0.0001)
  Scheduler:       LinearWarmup(5ep) + CosineAnnealingLR
  Grad accumulation: 4 steps (effective batch=16)
  Early stopping:  patience=25, min_epochs=25

  RESULTS (mean +/- std across 3 folds):
  ──────────────────────────────────────
  sensitivity     0.9074 +/- 0.0367
  specificity     0.9816 +/- 0.0029
  precision       0.9086 +/- 0.0089
  accuracy        0.9692 +/- 0.0063
  auc_roc         0.9918 +/- 0.0059
  f1              0.9076 +/- 0.0193
  ──────────────────────────────────────

  TIMING:
  Fold 1: 81 epochs, 141.6 min
  Fold 2: 76 epochs, 130.6 min
  Fold 3: 69 epochs, 118.6 min
  Total: 397.8 min

MODEL: VGG3D
  Architecture:    VGG3D
  Parameters:      0.87M (866,338)
  Input shape:     (1, 64, 64, 64) at 1mm isotropic
```

```
Optimizer:          AdamW (lr=0.0003, wd=0.0001)
Scheduler:          LinearWarmup(5ep) + CosineAnnealingLR
Grad accumulation: 4 steps (effective batch=16)
Early stopping:  patience=25, min_epochs=25

RESULTS (mean +/- std across 3 folds):
──────────────────────────────────────────
sensitivity      0.9243 +/- 0.0192
specificity      0.9831 +/- 0.0017
precision        0.9165 +/- 0.0041
accuracy         0.9733 +/- 0.0032
auc_roc          0.9944 +/- 0.0021
f1               0.9203 +/- 0.0095
──────────────────────────────────────────

TIMING:
Fold 1: 54 epochs, 58.0 min
```